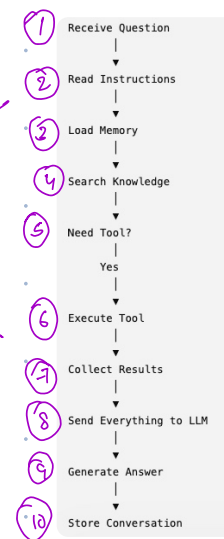
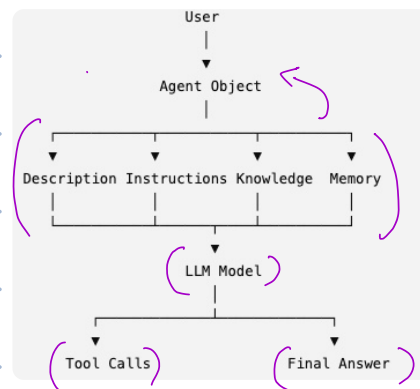


Agent → LLM → Reasoning →

Agent → Agent → Central element

Agent



Agent

① Model → LLM

- OpenAI
- Anthropic
- Google
- Groq
- Ollama
- Together
- Azure OpenAI

```
Agent(  
  model=OpenAIChat(id="gpt-4o")  
)
```

② Description

→ Model → Who it is

"You are Software architect"

③ Instruction

→ How the Agent should behave

```
instructions=[  
    "Always answer in Markdown.",  
    "Never guess.",  
    "Explain step by step."  
]
```

→ Be Concise

→ Use Bullet point

→

→ Explain

④ Markdown

(markdown = True)

With MD

```
# Python  
  
Python is a programming language.  
  
## Features  
  
- Easy  
- Fast  
- Popular
```

Without MD

Python is a programming language.

It is popular.

⑤ Show Tool calls

True / false

Search today's AI news.

→ Search tool
→ Answer

Tool Used:

DuckDuckGo Search

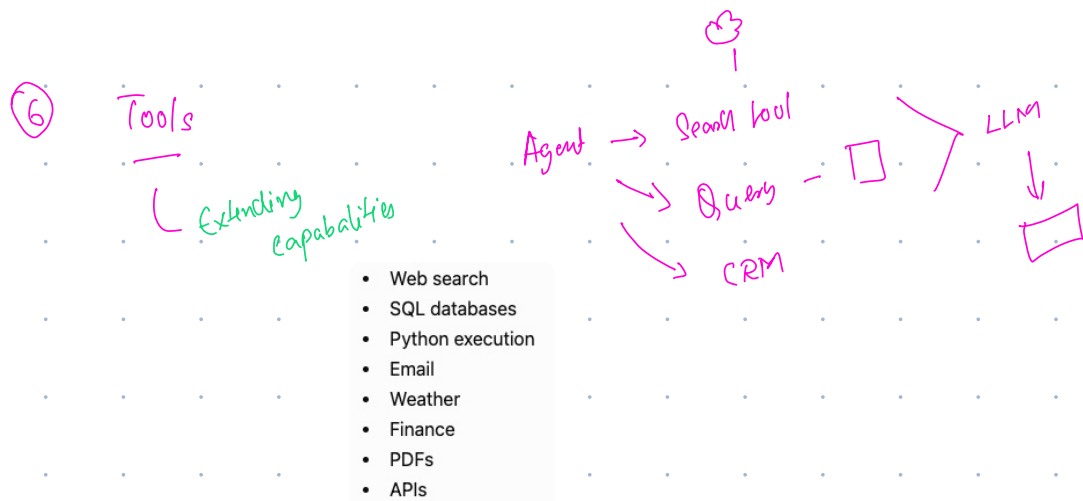
Searching...

Results Found...

Generating Summary...

> Tools → Answer

ⓧ



⑦ Knowledge → (Agent) → Reference Material
↳ RAG Pattern

Knowledge — information is available

8) Memory → previously, has conversation

My Name is Ahmed

Who am I
✓ Ahmed

Storage → Single execution

Memory → Storage — SQLite
Postgres

Reasoning

— Agent — additional efforts — Complex Task

1. Break the task into smaller parts.
2. Consider constraints.
3. Combine findings into a coherent plan.

Structured O/p

↳ Returning plaintext — plaintext
↳ JSON

```
{  
  "name": "John",  
  "age": 28,  
  "profession": "Engineer"  
}
```

Session ID

User A → Context — Session 123

User B → Context — Session 124

Agent

- |
- ├ Model
- ├ Description
- ├ Instructions
- ├ Markdown
- ├ Tools
- ├ Knowledge
- ├ Memory
- ├ Storage
- ├ Reasoning
- └ Structured Output

```

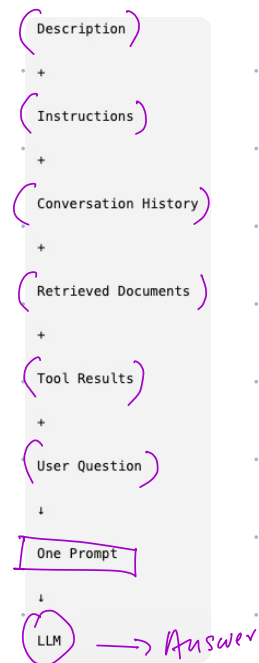
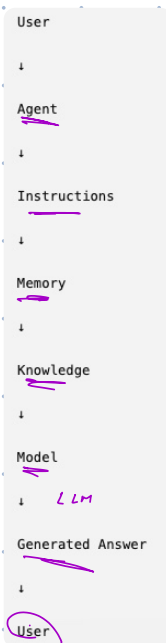
from phi.agent import Agent
from phi.model.openai import OpenAIChat
from phi.tools.duckduckgo import DuckDuckGo

agent = Agent(
    model=OpenAIChat(id="gpt-4o"),
    description="You are an AI research assistant.",
    instructions=[
        "Answer step by step.",
        "Use markdown.",
        "Never guess."
    ],
    markdown=True,
    tools=[DuckDuckGo()],
    show_tool_calls=True
)

agent.print_response(
    "Latest AI news today"
)

```

→ invoking



Aspect	Agno <i>CrewAI</i>	LangGraph
Philosophy	(Build agents quickly)	(Build reliable workflows)
Learning curve	<u>Easy</u>	<u>Moderate to steep</u>
Control	<u>High-level abstraction</u>	<u>Low-level control</u>
Multi-agent	<u>Built-in and simple</u>	<u>Powerful but requires more design</u>
Debugging	<u>Easier initially, harder internally</u>	<u>Easier because every step is explicit</u>
Production systems	<u>Good</u>	<u>Excellent</u>
Enterprise adoption	<u>Growing</u>	<u>Very strong</u>
Best for	<u>Startups, prototypes, internal tools</u>	<u>Banking, healthcare, enterprise AI</u>